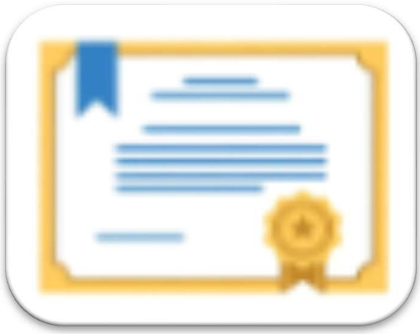
CERTIGEN: AUTOMATED CERTIFICATE GENERATION AND MANAGEMENT SYSTEM

PROJECT PROPOSAL

24MNC201



SUBMITTED BY: *Agnes Benoy*

ROLL NUMBER: 12

INSTITUTION: AMAL JYOTHI COLLEGE OF ENGINEERING KOOVAPPALIY

PROJECT TITLE: CertiGen – Automated Certificate Generation and Management System

TOOLS USED: Microsoft Excel, Microsoft Word, Mail Merge, and PDF Export

1, INTRODUCTION

In educational institutions, certificates are commonly issued to students to recognize their participation, achievements, assessment performance, and successful completion of workshops or other academic activities. Preparing these certificates manually can be time-consuming, particularly when a large number of students are involved. Each certificate may require the student's name, department, workshop details, assessment marks, and grade to be entered individually.

CertiGen – Automated Certificate Generation and Management System is a project designed to simplify and organize this process using Microsoft Excel and Microsoft Word. The system combines student data management, performance analysis, certificate template design, and automated document generation through the Mail Merge feature.

Microsoft Excel is used to maintain student records, organize assessment information, calculate or display results, and present data through analysis tables, charts, and PivotTables where applicable. Microsoft Word is used to design a professional certificate template containing predefined fields for student information. Mail Merge connects the Excel data source with the Word template and generates personalized certificates for multiple students.

The generated certificates can then be reviewed and exported into PDF format for convenient storage, printing, and sharing. The project demonstrates how commonly available office applications can be integrated to reduce repetitive work and maintain consistency in document preparation.

2. PROBLEM STATEMENT

Preparing certificates manually for multiple students involves repeatedly entering personal details, assessment marks, grades, department information, and workshop details into individual documents. This process requires considerable time and effort, especially when certificates must be prepared for an entire class or a large group of participants.

Manual certificate preparation can also lead to problems such as spelling mistakes, incorrect marks, inconsistent formatting, missing information, and differences in the layout of certificates. When student information changes, the corresponding certificates may need to be edited individually, increasing the workload and the possibility of errors.

Additionally, maintaining student performance information separately from certificate documents can make it difficult to organize records and review assessment results efficiently.

Therefore, a structured and automated approach is needed to organize student information, design a reusable certificate template, and generate multiple certificates with consistent formatting.

The proposed CertiGen system addresses this problem by using Microsoft Excel as the student data source and Microsoft Word as the certificate design platform. Through Mail Merge, information from the spreadsheet can be inserted into the appropriate positions in the certificate template, reducing repeated manual entry and improving consistency.

3. OBJECTIVES

The primary objective of CertiGen is to develop an organized and efficient certificate-generation workflow using Microsoft Excel and Microsoft Word.

The specific objectives are:

3.1 Student Data Management

To create and maintain a structured Excel worksheet containing student information, assessment marks, results, grades, and other details required for certificate preparation. Organizing the information in a tabular format makes it easier to update, review, and retrieve records.

3.2 Performance Analysis

To organize and analyze student performance information using Excel features such as formulas, summary tables, charts, and PivotTables, as included in the workbook. These features help present information in a structured and understandable manner.

3.3 Certificate Template Design

To design a professional and reusable certificate template in Microsoft Word. The template includes appropriate page formatting, borders, background design, a watermark where required, and clearly arranged fields for student details and achievement information.

3.4 Automated Certificate Generation

To use Microsoft Word's Mail Merge feature to connect the certificate template with the Excel student database. This enables the generation of multiple personalized certificates without manually creating each document from the beginning.

3.5 Result-Based Certificate Selection

To filter the recipient list according to student results when required. For example, records with a Result value of "Pass" can be selected for certificate generation, according to the project's specified criteria.

3.6 Consistency and Accuracy

To maintain a consistent design and document structure across all generated certificates while reducing repetitive typing and the risk of manual data-entry errors.

3.7 PDF Export and Document Protection

To export the completed certificates into PDF format for convenient viewing, printing, and sharing. Where required, editing restrictions can also be applied to the final Word document to help prevent unintended modifications.

3.8 Practical Application of Office Tools

To demonstrate how Microsoft Excel and Microsoft Word can work together to automate a repetitive administrative task using their built-in features.

4. TOOLS AND TECHNOLOGIES USED

The project uses the following tools and technologies.

4.1 Microsoft Excel

Microsoft Excel is used as the primary student data management and analysis tool.

Its functions in the project include:

- Creating a structured student database.
- Storing student names, department details, assessment marks, results, and grades.
- Organizing records into rows and columns with appropriate headings.

- Using formulas and functions for calculations and result processing, where implemented.
- Creating analysis tables and charts to summarize student performance.
- Using PivotTables and PivotCharts, where included, to summarize and interpret data.
- Acting as the data source for Microsoft Word's Mail Merge feature.

Excel provides a centralized location for maintaining student information, making the certificate-generation process more organized.

4.2 Microsoft Word

Microsoft Word is used to design the certificate template and produce the final documents.

Its functions include:

- Setting the page size to A4 and orientation to landscape, as required by the design.
- Applying margins, page borders, background colours, and a watermark where configured.
- Formatting the certificate title, student name, achievement details, and signature areas.
- Creating a reusable certificate layout.
- Inserting Mail Merge fields corresponding to the Excel column headings.
- Previewing the personalized certificates before generating the final documents.
- Applying editing restrictions where required by the assignment.

Word provides the document design and formatting features necessary to create readable and professional certificates.

4.3 Mail Merge

Mail Merge is the main automation feature used in CertiGen. It connects a Word document with a data source, such as an Excel worksheet, and uses the records to create personalized copies of the document.

In this project, the Word certificate serves as the main document, while the Excel Student Data worksheet acts as the data source.

The template contains fields such as:

- Title
- Student_Name
- Department
- Workshop
- Assessment_Marks
- Grade

During the merge process, these fields are replaced with the corresponding values from each selected student record. This makes it possible to generate multiple certificates using a single template.

4.4 PDF Export

PDF export is used to create a portable copy of the completed certificates. The PDF format helps preserve the document's layout and makes the certificates convenient to view, print, and share.

After exporting, the PDF is checked to confirm that the text, formatting, and page layout appear correctly.

4.5 Microsoft Office Integration

The project demonstrates integration between Excel and Word. Excel manages the records and analysis, while Word handles certificate design and document generation. Mail Merge connects the two applications, creating a coordinated workflow without requiring a separate database system or a custom software application.

5. PROPOSED WORKFLOW

The CertiGen project follows a sequence of steps, beginning with student data preparation and ending with the final certificate output.

Step 1: Student Data Collection and Entry

Student details are entered into a structured worksheet in Microsoft Excel. The worksheet contains the information required for certificate generation, including student names, department, workshop details, assessment marks, results, and grades.

Each column is assigned a clear heading so that the data can be identified correctly during the Mail Merge process. The records are reviewed to ensure that the information is complete and consistently entered.

Step 2: Data Organization and Formatting

The student records are organized into a tabular format. Headings are formatted, column widths are adjusted, borders are applied where appropriate, and the data is arranged for readability.

Formulas or functions may be used for calculations and result processing, depending on the workbook's implementation. The data is checked before it is used as the source for generating certificates.

Step 3: Student Performance Analysis

Excel's analysis features are used to summarize the available student data. Summary tables and charts help present assessment information in a visual format. PivotTables and related charts can be used to group and summarize the data where configured in the workbook.

The analysis worksheet provides a convenient way to review the information maintained in the student database.

Step 4: Certificate Template Creation

A certificate template is created in Microsoft Word. The page is configured according to the intended certificate design, including the page size, landscape orientation, margins, borders, and background formatting.

The certificate contains the necessary headings, student information areas, workshop details, assessment information, and signature spaces. Merge fields are positioned where the corresponding student details should appear.

The template is saved separately so that it can be reused for future certificate-generation tasks.

Step 5: Connecting Excel with Word

The Word certificate template is connected to the Excel workbook using the Mail Merge feature.

The user selects the Excel file as the recipient list and chooses the worksheet containing the student records. Word uses the worksheet's column headings to identify the available merge fields.

The field names in Word must correspond to the actual headings in Excel. This connection allows information from the spreadsheet to be inserted into the certificate automatically.

Step 6: Recipient Filtering

If the project requires certificates to be issued only to eligible students, the recipient list can be filtered according to the Result field.

For example, setting the condition to Result equals Pass selects records marked as Pass. This ensures that the merge uses the intended group of student records.

The recipient count should be checked after filtering to confirm that the correct records have been selected.

Step 7: Previewing the Certificates

The Preview Results feature in Word is used to inspect the personalized certificates before producing the final output.

The user checks whether the student names, department, workshop details, marks, and grades appear in the correct locations. Longer names and different record values are also reviewed to identify formatting problems.

Any missing fields, incorrect data, or layout issues are corrected before completing the merge.

Step 8: Generating the Certificates

After the preview is checked, the Finish & Merge feature is used to create the individual certificates.

Word generates a merged document containing a personalized certificate for each selected student record. Each certificate follows the same template design while displaying the appropriate information for that student.

The merged document is saved for further verification and final output preparation.

Step 9: PDF Export

The completed merged Word document is exported to PDF format. The PDF is opened and reviewed to verify that the certificates display correctly.

The checks include the number of pages, completeness of the student details, readability of the text, page layout, and visibility of the design elements. Any formatting issues are corrected in the Word document before exporting again.

Step 10: Final Verification and Submission

The final Word document and PDF are checked against the project requirements. Editing restrictions are applied to the completed Word document if required.

The project files are organized with clear filenames. The proposal, documentation, Excel workbook, certificate template, merged certificates, PDF output, screenshots, and demonstration video are prepared according to the instructor's submission requirements.

This completes the proposed workflow for CertiGen.

6. EXPECTED OUTCOMES

The expected outcomes of the CertiGen project are as follows.

6.1 Organized Student Records

The project provides a structured method for storing student information in Excel. The records can be reviewed and updated in a centralized worksheet, reducing the need to maintain separate manually typed lists for certificate preparation.

6.2 Efficient Certificate Generation

By using Mail Merge, multiple personalized certificates can be generated from a single Word template. This reduces the repetitive work involved in preparing each certificate individually and makes the process more efficient for larger groups of students.

6.3 Consistent Certificate Formatting

All generated certificates use the same underlying template. This helps maintain consistency in the page layout, fonts, borders, headings, and arrangement of student information.

6.4 Reduced Manual Data-Entry Errors

Since student details are drawn from the Excel worksheet, the need to retype information into each certificate is reduced. This can lower the risk of typing mistakes and inconsistencies, provided that the source data is accurate and the merge fields are configured correctly.

6.5 Improved Visibility of Student Performance

The Excel analysis tables, charts, and PivotTables included in the workbook help organize student performance information. These features make the available data easier to review and interpret.

6.6 Selective Certificate Generation

Filtering the Mail Merge recipient list allows certificates to be generated for a specified group of students, such as those whose results meet the project's eligibility criteria.

6.7 Portable Digital Output

The PDF output provides a convenient format for storing, printing, and sharing the completed certificates while helping preserve their intended appearance.

6.8 Reusable Certificate Template

The Word template can be reused with updated student records. For future batches, the data source can be updated and the Mail Merge process repeated, provided the required field headings and structure remain compatible.

6.9 Better Document Organization

The project produces a set of organized digital files, including the Excel workbook, Word template, merged certificate document, and PDF output. This makes the project easier to review, maintain, and demonstrate.

6.10 Practical Learning Outcome

The project demonstrates practical skills in spreadsheet organization, data analysis, document formatting, Mail Merge, PDF export, and basic document protection. It also illustrates how built-in office application features can be combined to solve a common administrative problem.

7. SCOPE OF THE PROJECT

The scope of CertiGen is limited to certificate preparation using Microsoft Excel and Microsoft Word. It covers student record organization, performance analysis within the workbook, certificate template design, Mail Merge, recipient filtering, document generation, PDF export, and final document checks.

The current project does not include a dedicated web application, an online student portal, a cloud database, automatic online certificate verification, or a custom certificate-validation system. These features are outside the scope of the described implementation.

The system's accuracy depends on the correctness of the source records, the consistency of the spreadsheet headings, and the proper configuration of the Mail Merge fields.

8. CONCLUSION

CertiGen – Automated Certificate Generation and Management System demonstrates how Microsoft Excel and Microsoft Word can be integrated to simplify the preparation of certificates for multiple students.

Excel provides the structured data source and supports the analysis of student performance. Word provides a reusable certificate design, while Mail Merge connects the two applications to generate personalized documents. PDF export makes the completed certificates convenient to store, print, and share.

By reducing repeated manual entry and maintaining a consistent certificate layout, the project offers a practical approach to improving the efficiency and organization of certificate preparation. It also provides valuable hands-on experience in spreadsheet management, document design, office automation, and digital document handling.

Overall, CertiGen demonstrates that existing office tools can be combined effectively to automate a repetitive administrative task without the need to develop a separate software application.